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房地產及建設系



Smart QS Hackathon 2026 - Workshop 5

From Dumb Points to Smart As-Built Quantities

Prof Frank Xue
Associate Professor
Department of Real Estate and Construction
The University of Hong Kong
xuef@hku.hk
<https://frankxue.com>



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Learning outcomes

◦ Upon the completion of this session, you should be able to

- 1) **Name** LiDAR scan and an editing tool
- 2) **Compare** manual editing functions
- 3) **Measure** 3D points and surfaces
- 4) **Cluster** points to objects
- 5) **(Re-)train** deep learning model

Evidence

Means

Quantity

Semantics

Frank's legend of level

Acknowledge (Level 1)

Analysis (Level 2)

Application (Level 3)



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Part I. Basics of point clouds

(L1, L2)



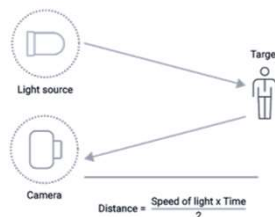
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I.1 From theodolite to laser scan

- Surveying tools evolution
 - Theodolite (1787): angle meter
 - Total station (1971): electronic optical distance meter (EDM) + electronic angle meter
 - Laser scanner (1993): Laser time-of-flight (TOF); Phase-shift (PS); auto rotating mirror
 - Color laser scanner (2018): + color imagery



(Sources: Wikimedia, CC-BY-SA; Leica)



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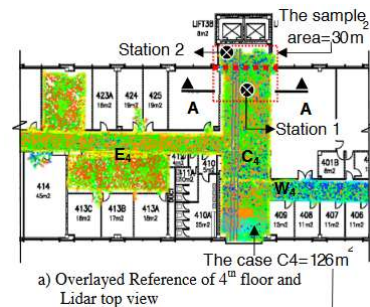
Sepasgozar, S. M. E., Lim, S., Shirowzhan, S., & Kim, Y. M. (2014). Implementation of as-built information modelling using mobile and terrestrial lidar systems. In *ISARC. Proceedings of the International Symposium on Automation and Robotics in Construction* (Vol. 31, p. 1). IAARC.

I.1 Terrestrial versus mobile LiDAR

Characteristics	Mobile lidar	Terrestrial lidar
Operating	Without Adjustment	With Adjustment
Portability	510 g	7.6 kg
Required skill	Low	High
Estimated cost	About \$25,000	About \$60,000
Resolution	Medium	Very high
Accuracy	Millimetre	Millimetre
Spatial data Retrieval	Semi-automatic	Semi-automatic
Spatial data accuracy	±5 to 30 millimetre	± 3 to 12 millimetre
Spatial data speed	Real time retrieval	Real time retrieval
Range distance	20	1000
Operation time (for the sample)	10 minutes	30 minutes



- A corridor scan study by Sepasgozar et al. (2014)
 - mobile LiDAR is faster and easier,
 - terrestrial LiDAR is more accurate and higher-resolution

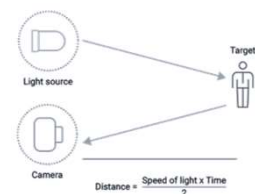
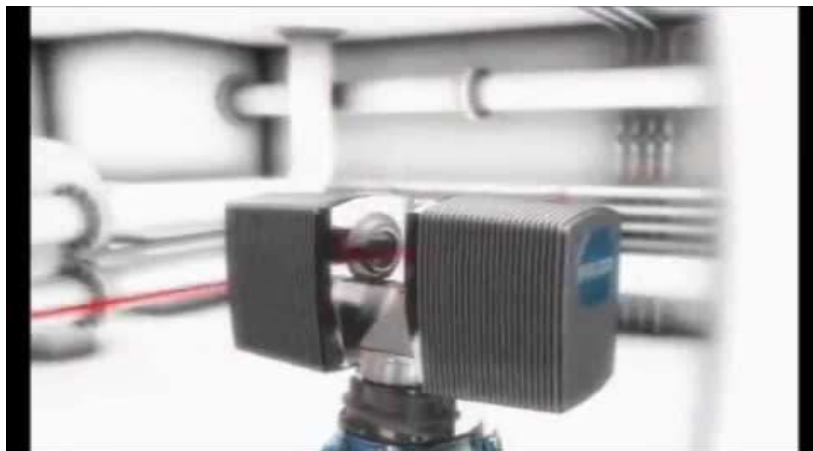


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I.1 LiDAR: Light Detection And Ranging



Astrolabe Engineering. (2012). How Laser Scanning Works v2, Youtube video (CC-BY)
<https://www.youtube.com/watch?v=jNwpqXtIM-E>



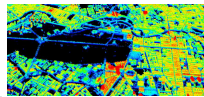
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I.1 A summary of 3D survey of buildings

◦ SAR [PS] 合成孔徑雷達



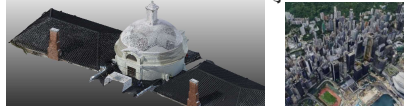
Bellagio Hotel, Las Vega
(Zhu & Bamler 2014)

◦ LiDAR 激光雷達

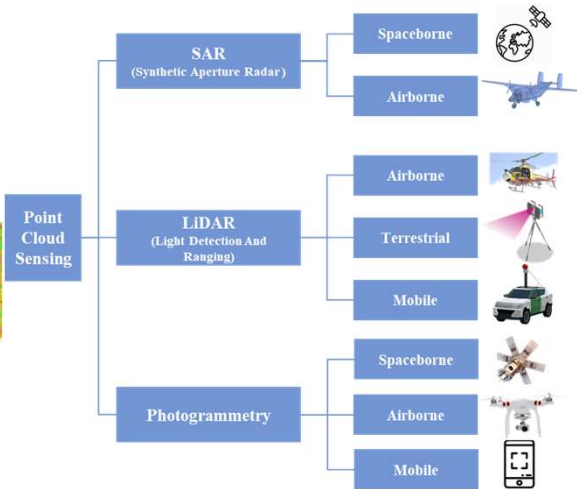


Furniture (Xue et al. 2019c), Scan-to-BIM
Challenge, HKU Campus (Xue et al. 2019f)

◦ Photogrammetry 攝影測量學



Building rooftop (Xue et al. 2019d), city model (Li et al. 2022)



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I.1 Some use cases of 3D surveying data

◦ SAR 合成孔徑雷達

- mm-accuracy
- Coverage



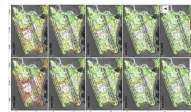
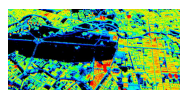
◦ LiDAR 激光雷達

- mm/cm/dm
- No distortion
- Intensity

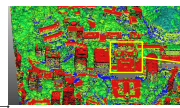


◦ Photogrammetry 攝影測量

- cm-accuracy
- Colorful
- Cheaper



Ground settlement, building deformation (Wu et al. 2020a; 2020b)



Roof albedo (Xue et al. 2019f), indoor CFD simulation (Source: Author, 2022)



Kowloon Wall City 3D model
(Source: patrick-sketchfab.com)

HKU @MineCraft (Source: Author, 2021)



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I.2 Point clouds

- Point
 - A **location** in space, 0D (no width, length, or thickness)
 - Structured format: {x, y, z}, [R, G, B, Nx, Ny, Nz, Cls, Int., ID...]
- Cloud
 - An **unstructured collection** [of water droplets or ice crystals]
 - Dense when looking at a distance, sparse closely
- Point cloud (PC)



A point cloud of HKU Campus (Source: Author, 2019)



A close look of cloud at Mount Hua (Source: Author)



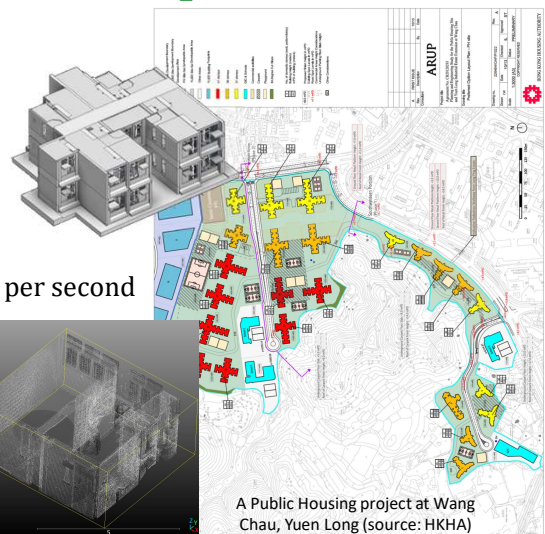
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I.1 PC is an unstructured, as-built sample data

- A drawing is a decision
(↕ lifecycle/dimension gap ↕)
- A BIM object represent a commitment
(↕ semantic gap ↕)
- A cloud of points is only **evidence**
 - Millions of 3D points where a scanner received per second
 - No walls, no columns, no payment items
 - until we segment/measure/detect structure.



A Public Housing project at Wang Chau, Yuen Long (source: HKHA)



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I.2 PCD processing and editing

- Like editing 2D selfies
 - Geometry/x,y,z, color
 - Normal, scalars, strength
 - Cropping, segmentation, slicing
 - Transformation
 - Fine registration
 - Sampling (down-/up-)
 - Measurement / local plane
 - Labeling / type / object ID
 - Coarse pose estimation
 - Coloring
 - Videoing
- (Object-free)
- (Object oriented)



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I.2 Two methods for PCD collection



LiDAR scan of Main Building (partial)
(Source: you guys, 2023)



Photogrammetry mesh of HKU University lodge
(Source: author, 2020)



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I.2.1 Examples of manual as-built modeling



kubitTilo. (2010). From 3D Laser Scanning to CAD models, Youtube video (CC-BY)
<https://www.youtube.com/watch?v=VloiTu2n0JE>

- Manual modeling works well
 - Though costly
 - Time
 - Human resources

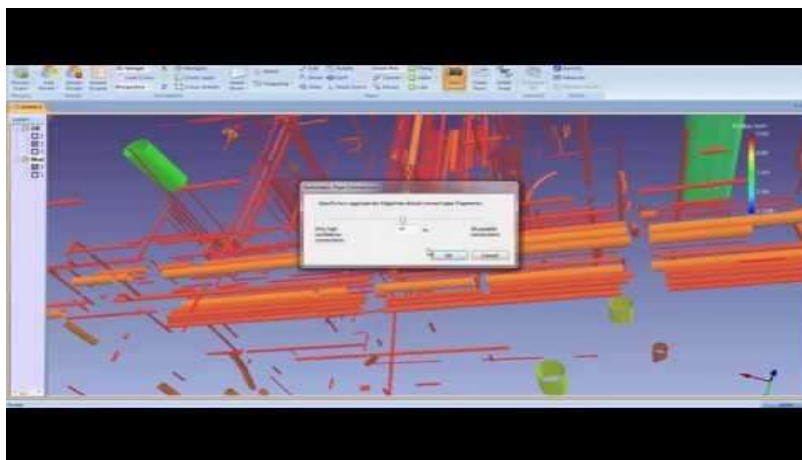


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I.2.1 Examples of semi-auto modeling of pipes



ClearEdge3D. (2015). EdgeWise Overview Video, Youtube video (CC-BY) <https://www.youtube.com/watch?v=ma4IyaWR4ko>

- Ready-to-use add-ons
 - Verity/EdgeWise
 - Automatic, semi-automatic

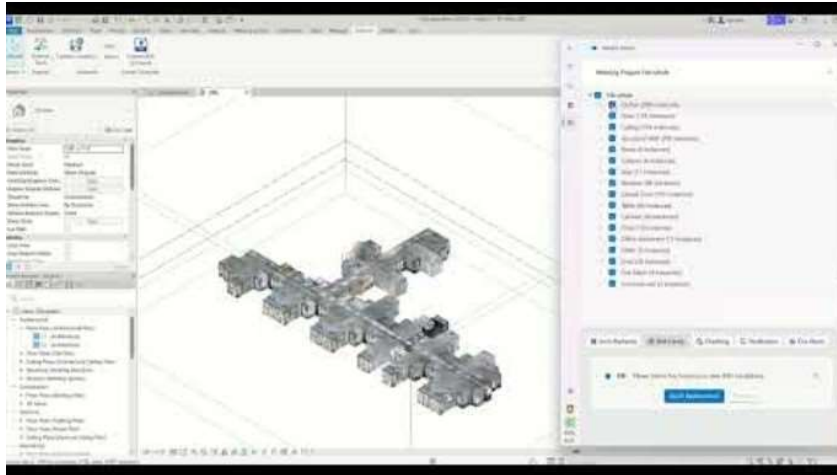


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I.3 Example of semi-automatic indoor modeling



- Faster modeling with auto-segmentation addon
- Works like Photoshop layers
- Thanks to Hank

Scan-to-BIM test. Source: Authors (CC-BY)
<https://www.youtube.com/watch?v=k33u08wTnR4>



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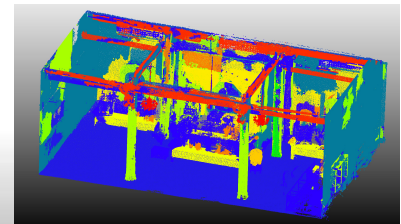
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I.2.2 Heritage BIM/PCD archive

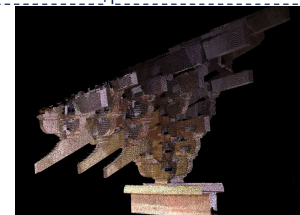


PCD preview

Semantic segmentation



Components





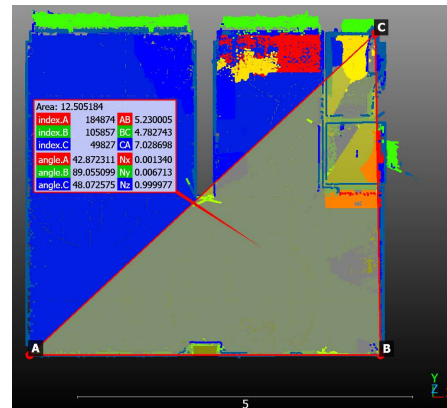
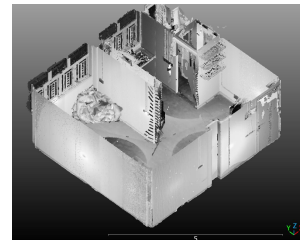
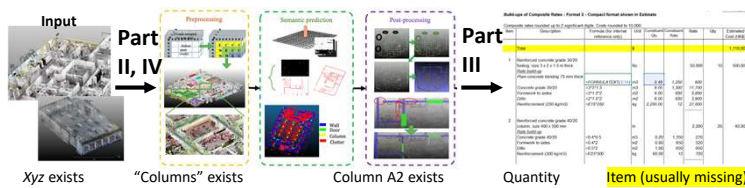
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I.2.3 Measurement

- Quantity/Payment is NOT in PCD;
 - identify progress and
 - assess amounts due.
 - A QS still has to name the item, unit, and rules.
- Change “head” for smart QS



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Part II. Point cloud editing

(L3)





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Exercise 1: Segment and register points



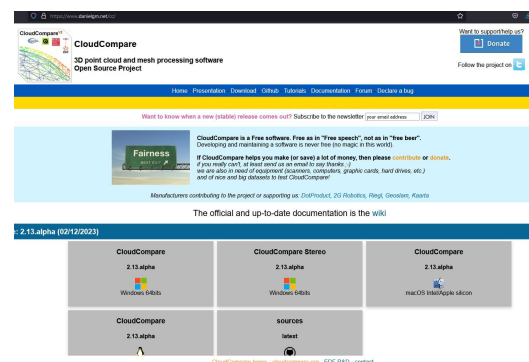
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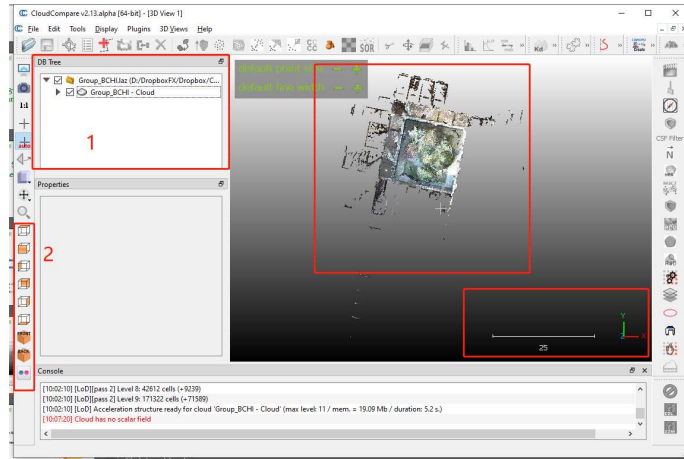
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II.1 Software and data

- CloudCompare
 - Open source
 - Download latest version
 - <https://www.danielgm.net/cc/>
 - Windows, MacOS, Linux
- Data
 - A garden of Main Building, HKU
 - https://www.dropbox.com/s/ghe6essz5qbdpct/Group_BCHI.laz?dl=0
 - 14.7 million points
 - 54 MB (.laz compressed format)
 - Or 217 MB (.e57 format)



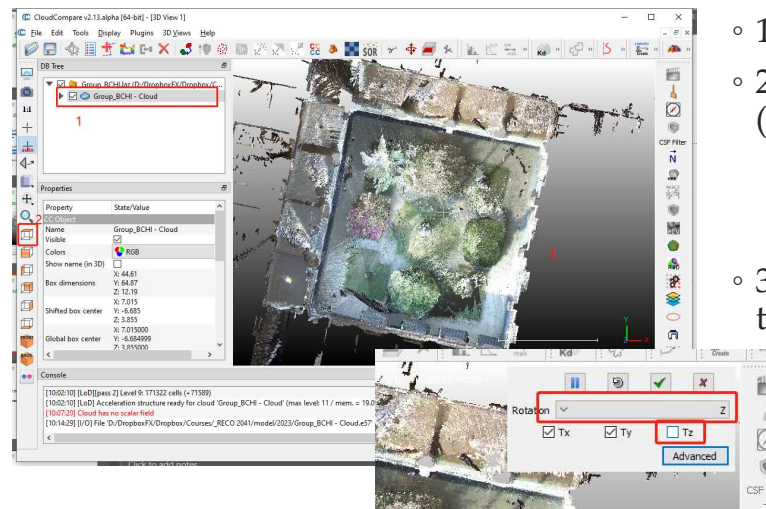
II.1 Viewing a point cloud



◦ Interface

- 1. Active target (tree of files and 3D objects)
- 2. View options
 - Screen
 - Other 3D device
- 3. 3D view
- 4. scale and 3D axes

II.1 Rotating to rough north



- 1. select target cloud
- 2. zoom in to target lines (parallel with the grid)
 - Use the Top view
 - Mouse scroll to zoom in/out
 - Hold right mouse key to pan
- 3. Menu – edit – translate/rotate
 - Limit Rotation to Z
 - Limit translation to X, Y

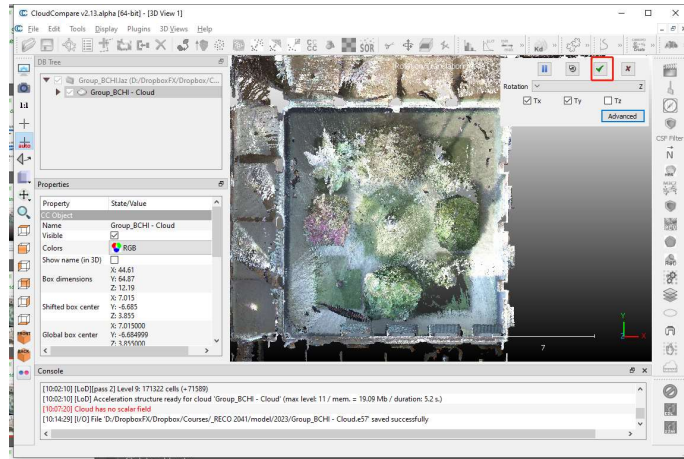


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II.1 Rotating to rough north



- 4. Hold mouse left key to rotate
- 5. hold mouse right key to pan
- 6. Let the corridor walls aligned to window border
- 7. Press check button to complete

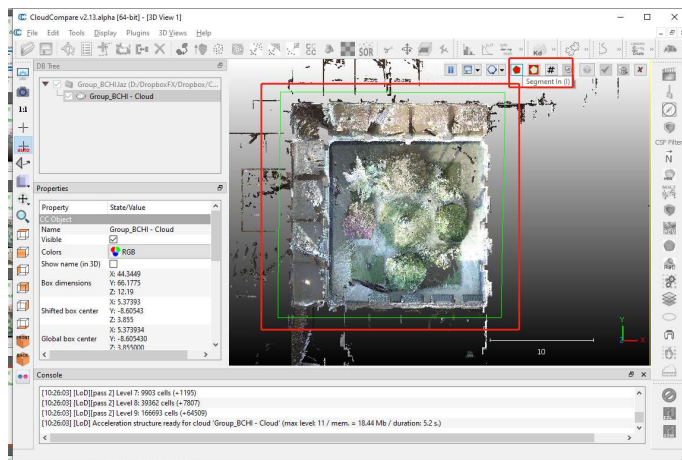


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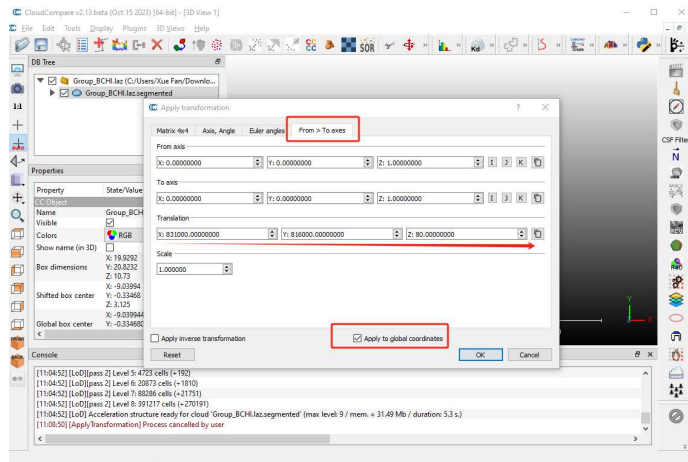
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II.1 Segment the garden (remove far-away noises)



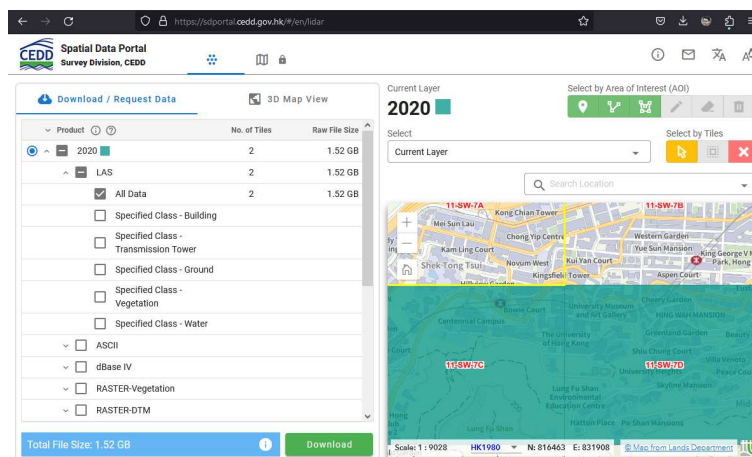
- 1. Zoom out a bit
- 2. Select the target cloud
- 3. Menu – Edit – segment
- 4. Left click to draw a polygon to enclose
- 5. Right click to end polygon
- 6. Click “Segment in” button
- 7. Click check button to confirm.

II.A [LS] Geo-registration to HKGS1980



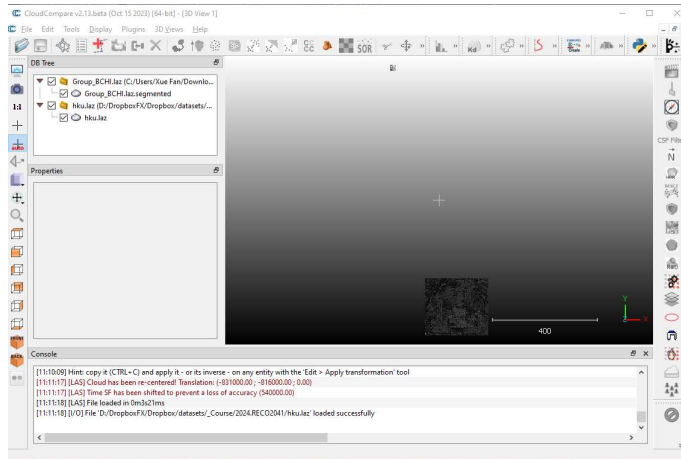
- Select the segmented cloud
- Menu-Edit-Applied Transformation
- Tab 4--“From >to axes”
- Translations:
 - X = 831000
 - Y = 816000
 - Z = 80
 - 31000m Easting, 16000m Northing from origin
- Check “Apply to Global”
- Press “OK”, then “Yes”

II.A [LS] Geo-registration to HKGS1980

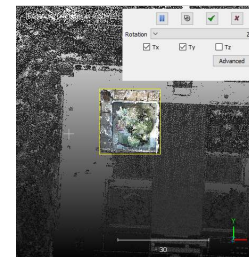


- Collect reference LiDAR cloud
 - <https://sdportal.cedd.gov.hk/#/en/lidar>
 - By CEDD, 2020
 - Territory-wide
- A “cooked” hku.laz for your download:
 - <https://www.dropbox.com/scl/fi/9c5sdxxvokgdxt8ummc31w/hku.laz?rlkey=ev6buhwfsfkd8mglwk7urlb0&dl=0>

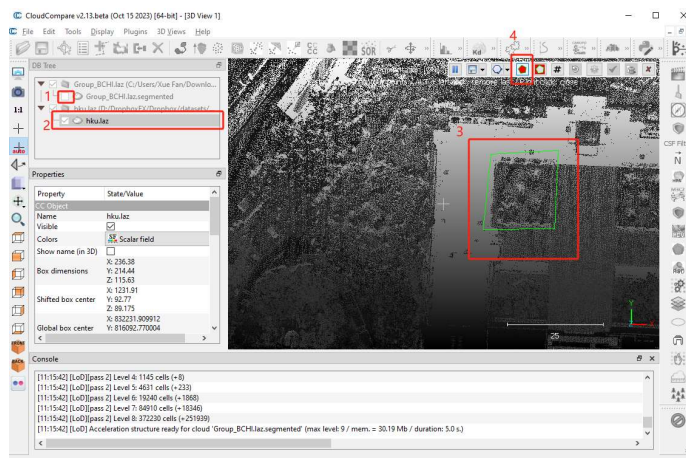
II.A [LS] Geo-registration to HKGS1980



- Load reference LiDAR “hku.laz”
 - Two clouds
 - About 200m distance
- Select our data, redo menu “Edit” - “Translate”

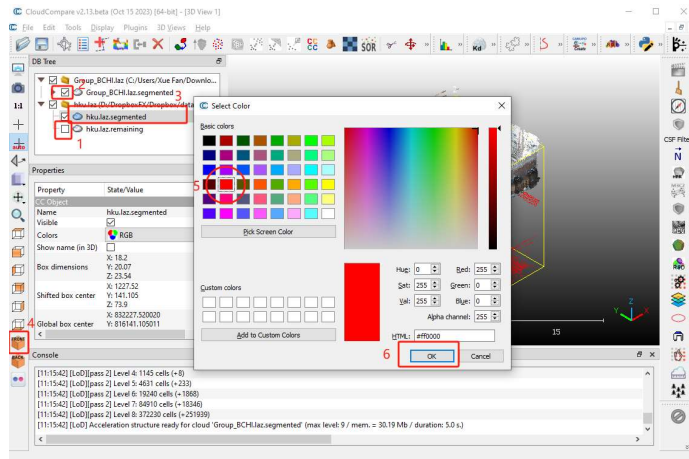


II.A [LS] Geo-registration to HKGS1980



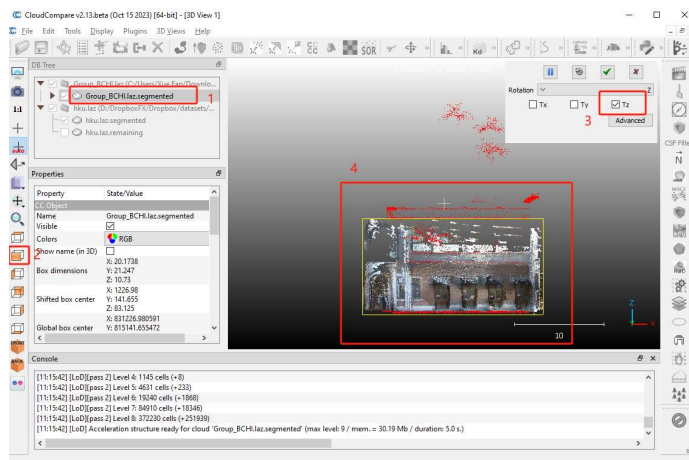
- 1 Unselect our scan
- 2 select reference cloud
- 3 Menu-“Edit”-Segment
 - Select the mini garden
- 4 use button “segment in”
- 5 press “check” button

II.A [LS] Geo-registration to HKGS1980



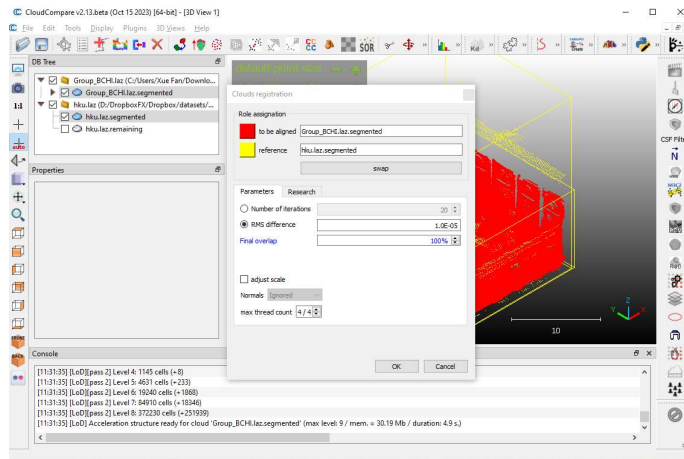
- View the two clouds
- 1 – select our cloud
- 2 – hide the remainder
- 3 – ensure current cloud is the reference
- 4 – change to front corner view
- 5 – Menu “Edit-Colors-Set Unique”
- 6 – choose RED.

II.A [LS] Geo-registration to HKGS1980



- 1 select our cloud
- 2 change to side view
- 3 Menu edit-translate/rot – restrict to Tz “z-trans”
- 4 Right mouse drag to move down to meet the ground level
- Note: change to other views to make sure it is roughly registered.

II.A [LS] Geo-registration to HKGS1980



- Accurate fine geo-registration with ICP
- Menu “Tool”-registration-Fine reg (ICP)
 - To be aligned: ours
 - Reference: the CEDD’s data
- Press “OK”



Exercise 2: 3D measurement and object creation (manual, preliminary)

(L3)

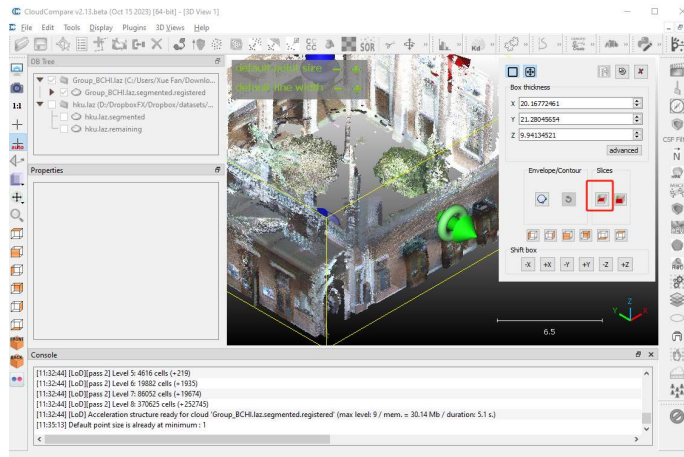


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II.2 ground removal



- 1. Hide all other parts
- 2. select the target cloud
- 3. Change view (bottom)
- 4. menu – tools – segment -- cross section
 - Drag blue arrow (not the circle)
 - Change views to ensure ground is invisible
- 5. click “slice” button to confirm

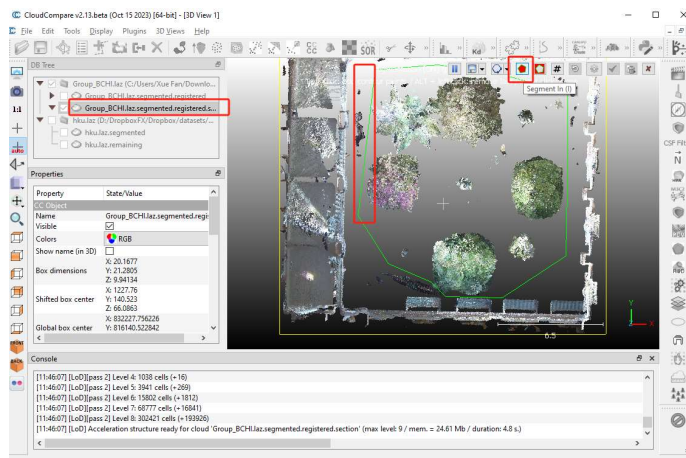


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II.2 vegetation/building segmentation



- 1. Select no-ground cloud
- 2. Menu edit – segment
- 3. Draw a polygon to encircle the trees and grass
- 4. Click “Segment in”
- 5. Confirm

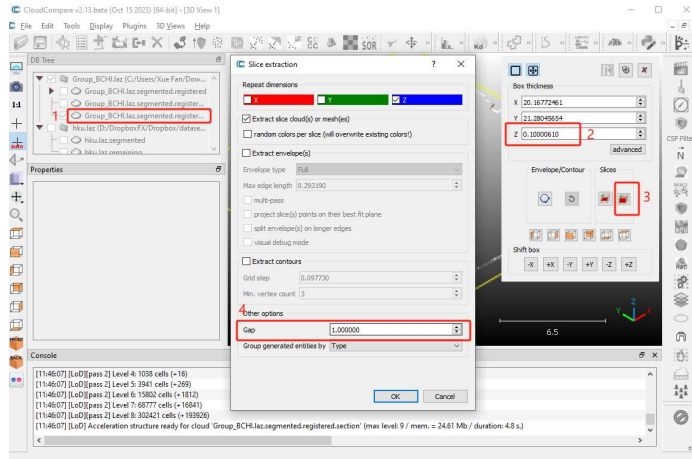


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II.2 Repeated z-slices for 2D drawing



- select cloud of *building*
- 1. Menu – tools – segment – cross section
- 2. set Height = 0.1 m
- 3. Click multiple slice
- 4. click “OK”
 - A gap = 1 m is also OK

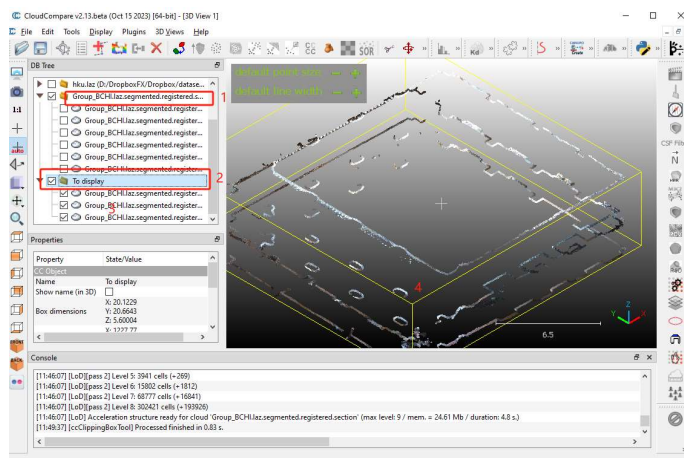


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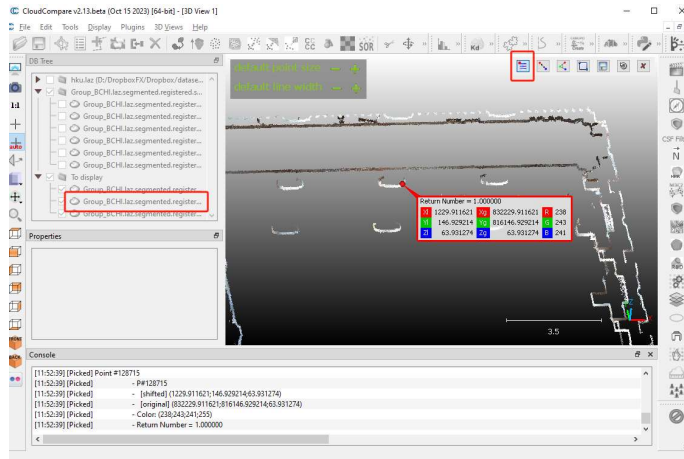
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II.2 Selection of z-slices



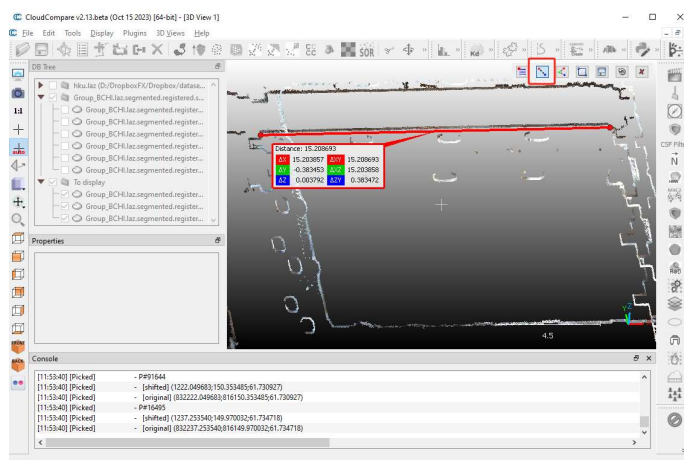
- Too many slices
- 1. Right click the base group, add an empty group
- 2. Rename the group
 - “To display” or any name
- 3. Drag a few slices to the new group
- 4. Uncheck the slides group to preview

II.2 3D measurement - point



- 1. Select one slice
- 2. Menu tools – point picking
- 3. click a point
 - X, y, z (HKGS80)
 - R, G, B

II.2 3D measurement - line



- Change to two-point mode
- Click two points in sequence
- A line
 - Length
 - $\Delta x, \Delta y, \Delta z$
 - $\Delta z = 0$ (horizontal)
 - $\Delta y = 0.1$ (almost aligned to x)
 - $\Delta xy, \Delta xz, \Delta yz$

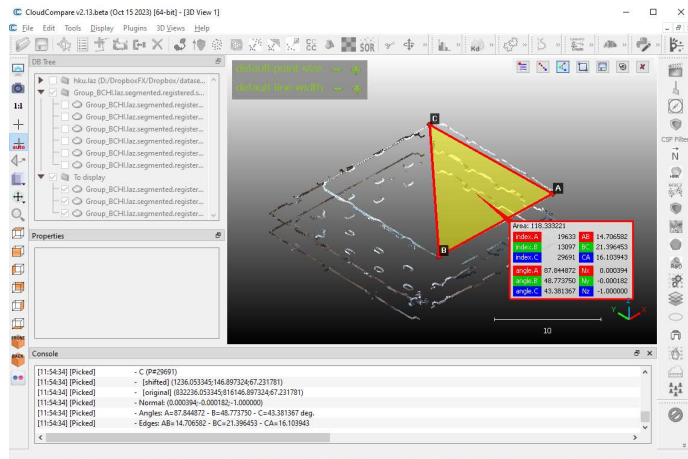


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II.2 3D measurement - area



- Change to 3-point mode

- Click 3 points to display
– areas



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Exercise 3: 3D object creation (preliminary; manual)

(L3)

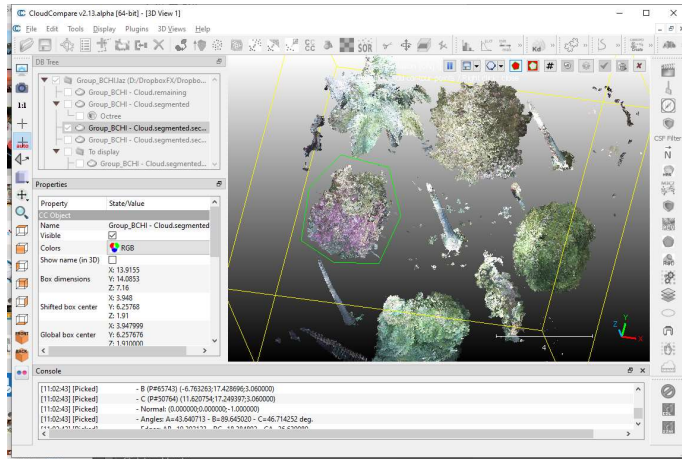


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II.3 Object segmentation



- Select the **vegetation** cloud
- Segment a flourished bush

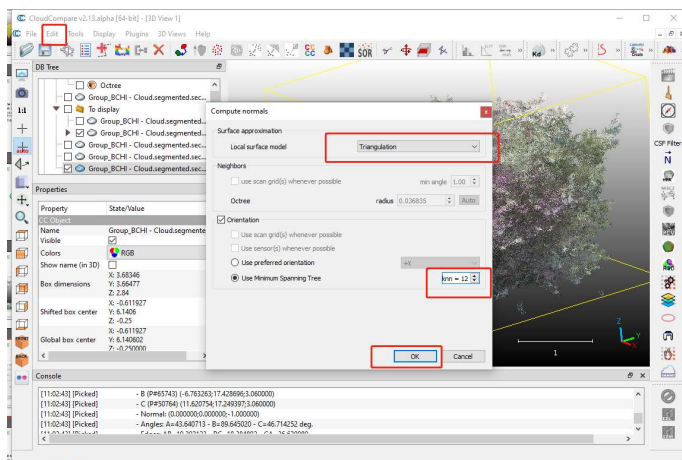


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II.3 Mesh creation - normal



1. Edit – normal – compute
 2. Use “Triangulation” surface mode
 3. set spanning tree k=12
- Click OK

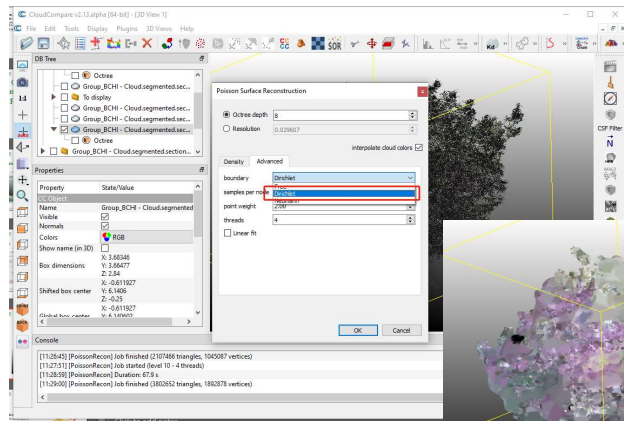


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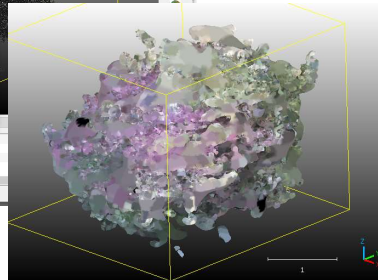


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II.3 Mesh creation - meshing



- Menu – plugins – PoissonRecon
- Go to “advanced”
- Change boundary to “Dirichlet”
 - For closer boundary
- Press OK
- Wait for 20 s
- You can uncheck normal for lighter colors

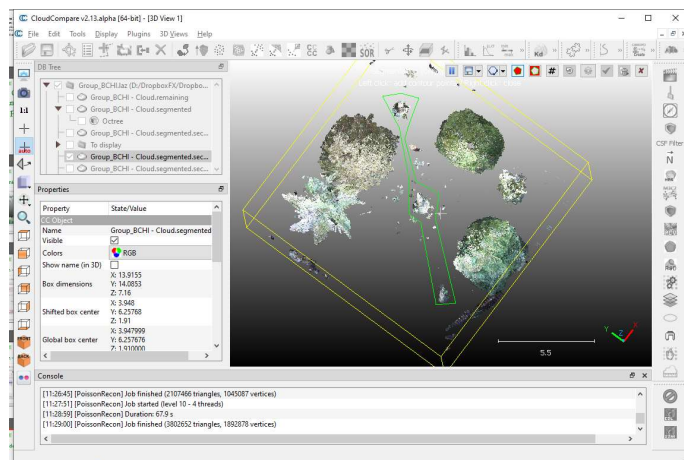


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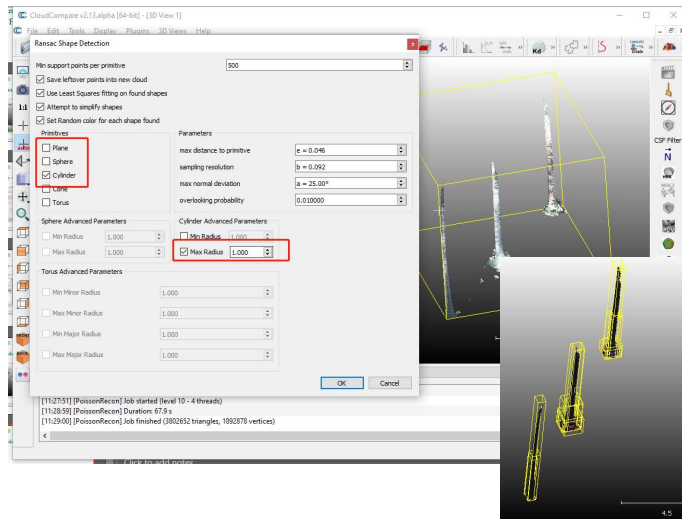
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II.3 Objects (cylinder detection)



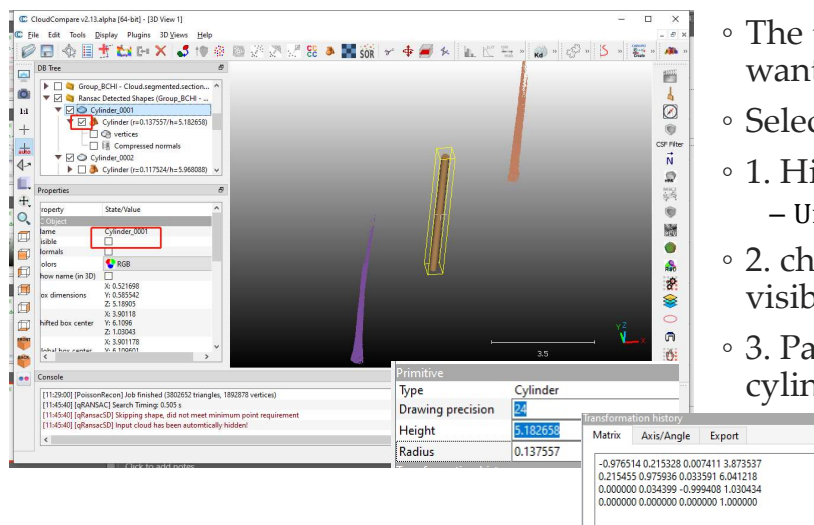
- 1. select remaining vegetation points
- 2. Edit – segment
- 3. Segment the trunks of 3 palm trees
- 4. Press check button

II.3 Objects (cylinder detection)



- Select the segmented
 - Menu – plugin – RANSAC shape detection
 - Limit to cylinder
 - Max radius = 1m
- Output:
- 17 cylinders

II.3 Checking result cylinders



- The top 3 are what we want
- Select No. 1
- 1. Hide support points
 - Uncheck “visible”
- 2. check cylinder’s visibility
- 3. Parameters of the cylinder

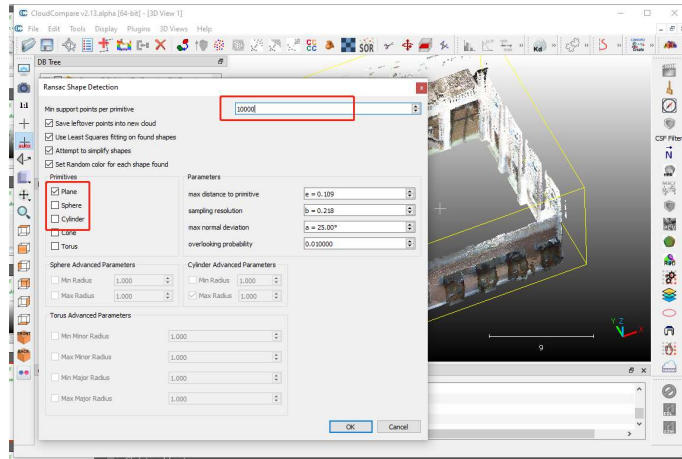


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II.3 Plane detection (walls)



- We select the corridor points **before** slicing
- Menu – plugin – RANSAC Shape Det ...
- 1. limited to “plane”
- 2. Set minimum support to “10000”

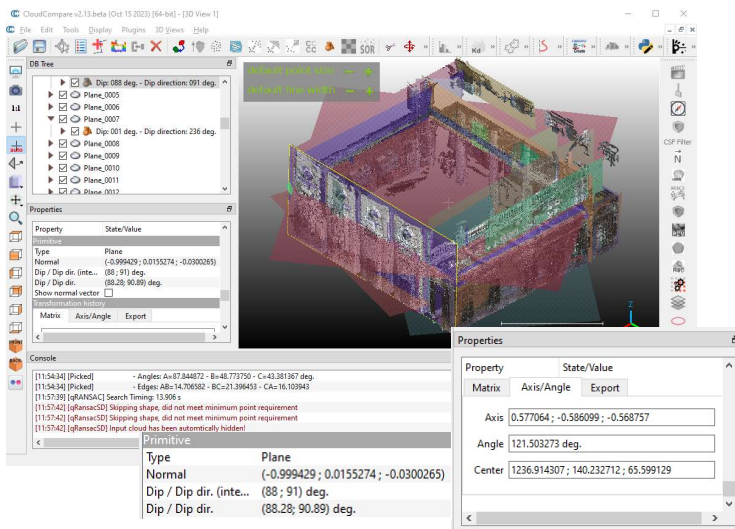


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II.3 Output planes



- 51 planes detected
- Select the first one
- It is a corridor wall
- Directions:
 - 90.89 degree
 - 0.89 degree from 90, for magnetic north, not polar north
 - Center of plane
 - (1236.91, 140.23, 65.60)

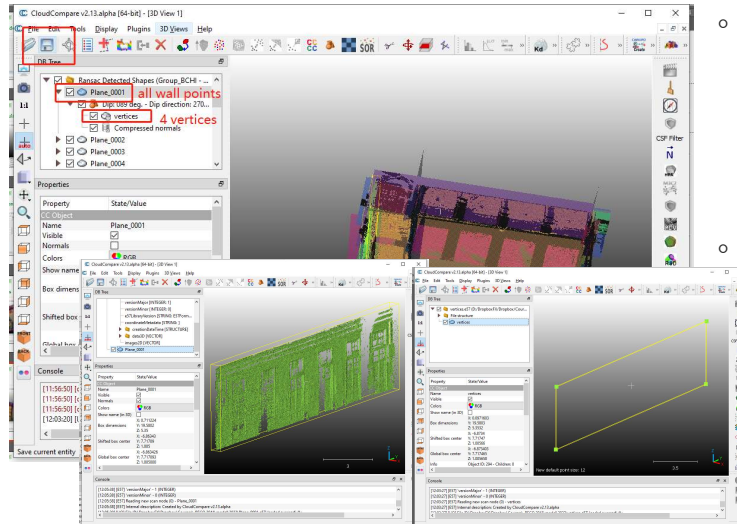


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II.3 Output points on the plane



- Save plane with points
 - “Save” button
 - Format: **e57** (has plane structures)
 - Can be imported by ReCap
 - Then Revit
- Save plane vertex
 - The “vertices” cloud
 - Number = 4



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II.4 Reflection

- Well done
 - You can edit and measure 3D points now.
- You can later import planar walls into Revit
 - This time, points not on the wall are excluded
- After-workshop thinking:
 - Regarding quantity/payment in QS, what are the pros and cons of referring and not referring to designed BIM?
 - Do you want full Scan-to-BIM or halfway (semantic labelled)





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Part III. Auto cluster 3D objects for Quantity

(L2, L3)



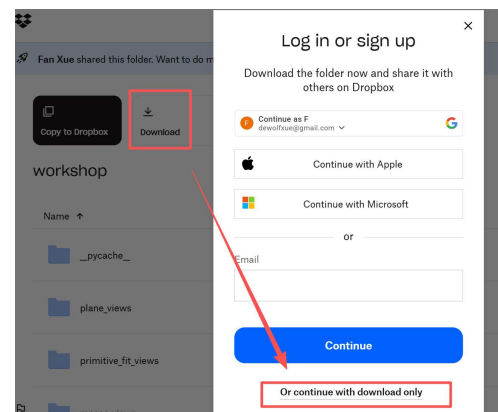
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III. Software and data

- Python 3 + numpy (mainstream versions)
- Data
 - https://www.dropbox.com/scl/fo/3pq1s13l9aifbc09i4pml/ACNz8Ag2R1Ey_cthnJuLUKE?rlkey=mlzrx1b6o86o8bwhkcswakid0&dl=0
 - 10 + 10 PCD files, a few clustering algorithms
 - 66 MB





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Xue, F., Lu, W., Chen, K., & Zetkovic, A. (2019). From semantic segmentation to semantic registration: Derivative-free optimization-based approach for automatic generation of semantically rich as-built building information models from 3D point clouds. *Journal of Computing in Civil Engineering*, 33(4): 04019024

III.1 Fitting/counting fixed size items



- Some items are fixed in size
 - Furniture, fire bell
- Fitting 3D to PCD is automatic
 - Left: IKEA Chair, no segmentation
 - Next page: PH fire bell, segmented
- Means
 - Least 3D fitting error,
 - Consider color weighting



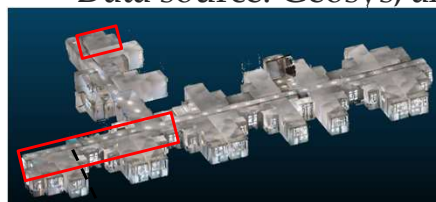
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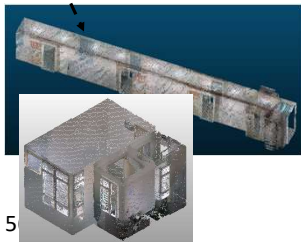
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III.1 Case of public housing, Yip Wong Road P1

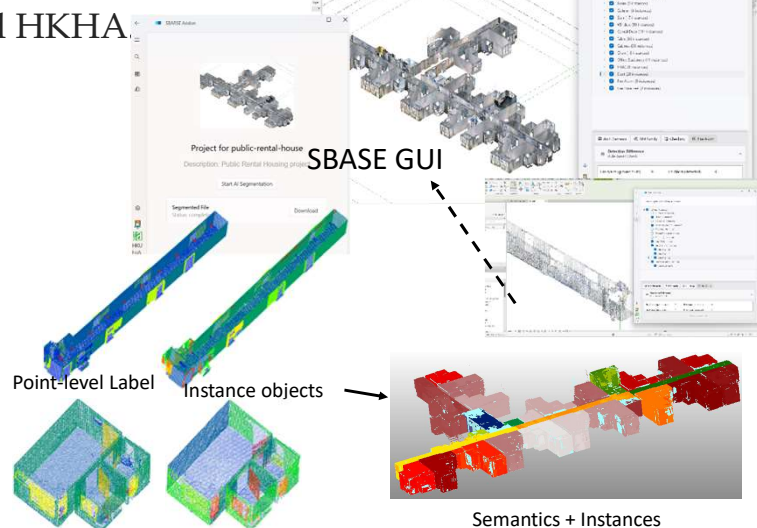
- Data source: Geosys, and HKHA



41,000,000 points in 2.5GB file



5



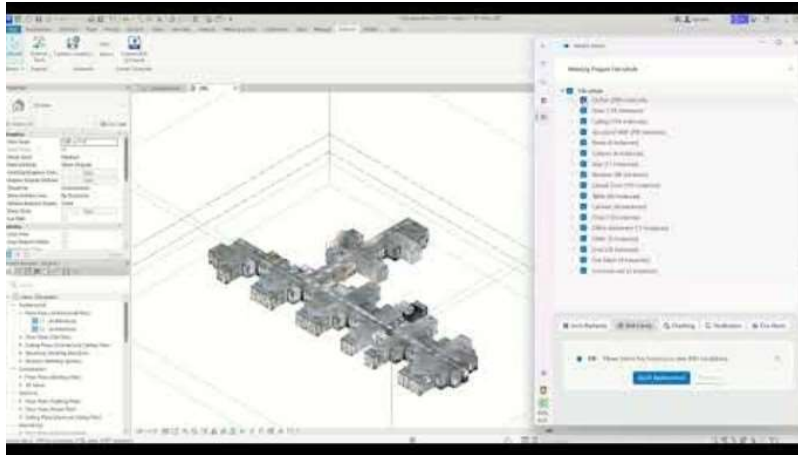


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III.1 Fitting/counting fixed size items (cont.)



- In segmented, choose fire
 - Confirm;
 - mark any errors
- Payment = unit cost * success
- Issus = unsuccessful

57

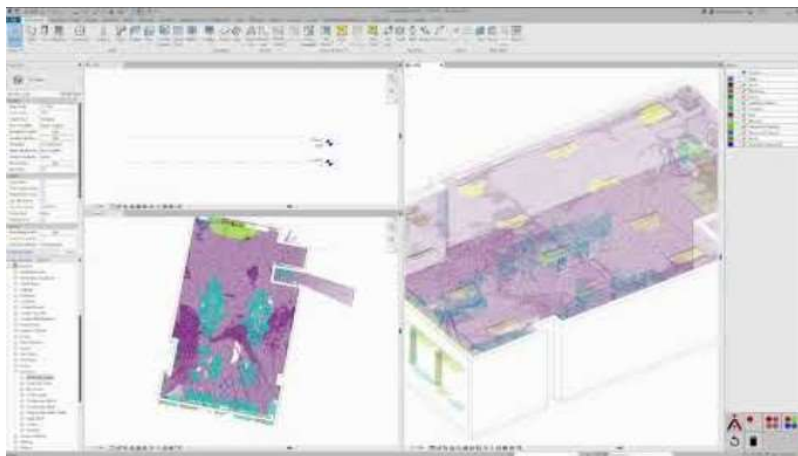


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III.1 Or drag BIM family/ visually confirm



- Another tool
 - Aurivas
 - Drag family, confirm
- 3D model is not a must for payment.
 - When quantity/ evidence is solid



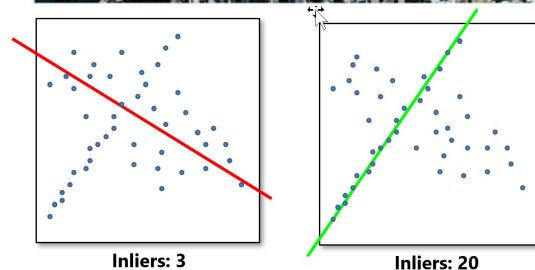
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III.2 Architectural elements

- Planes, cuboid, cylinder, ...
 - Not fixed dimension
- RANSAC (RANDOM Sample Consensus) in 4 steps:
 - Hypothesize (Pick a Random Sample)
 - Model (Associate A to B)
 - Test & count supports
 - Update the Best one
 - Until 100 or 1,000 runs.
 - Very fast algorithm
 - Let's test it using PCD



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III.2 Simplest RANSAC for planes

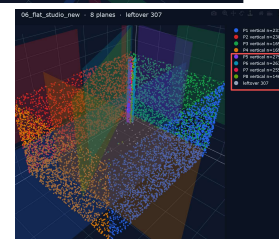
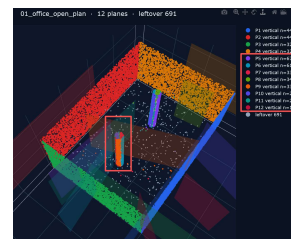
- Repeat:
 - RANSAC a plane (any 3 points), loop for 1,000 times,
 - Remove nest found plane
 - Until no sufficient planes or left over
- “simple_plane_ransac.py”

```

67 def peel_planes(pts: np.ndarray):
68     rng = np.random.default_rng(SEED)
69     leftover = pts.copy()
70     planes = []
71     for i in range(MAX_PLANES):
72         hit = ransac_plane(leftover, rng)
73         if hit is None:
74             break # until no more planes
75         nrm, d, idx = hit
76         inliers = leftover[idx]
77         span = inliers.max(0) - inliers.min(0)
78         tilt = float(np.degrees(np.arccos(np.clip(abs(nrm[2]), 0, 1))))
79         kind = "horizontal" if abs(nrm[2]) > 0.9 else "vertical" if abs

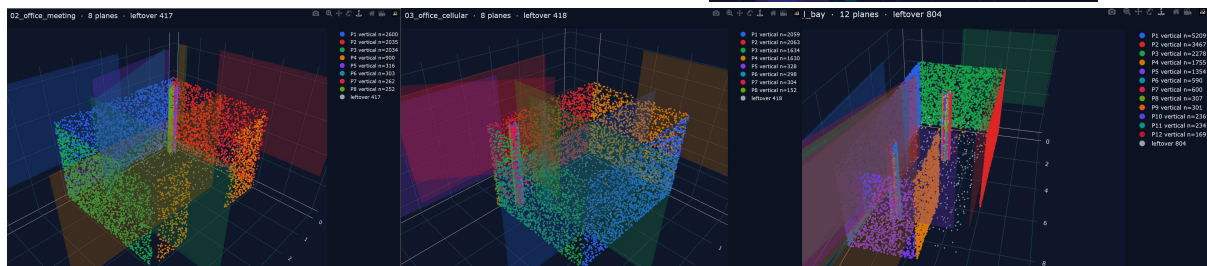
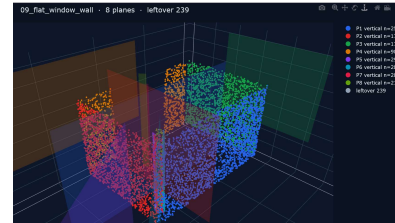
```

>Python simple_plane_ransac.py



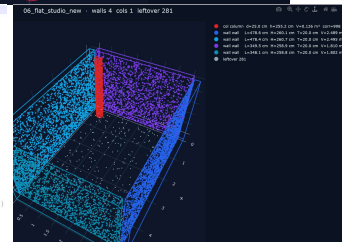
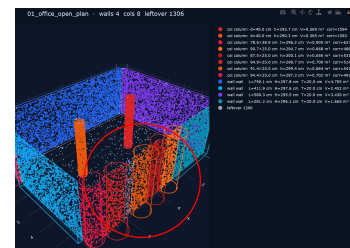
III.2 Simplest RANSAC for planes (cont.)

- “>python view_simple_planes.py”
- Tested on 10 synthetic scenes
 - “ransac_views”
 - Pros
 - Cons



III.2 RANSAC for wall cuboids and cylinders

- Now, repeat:
 - Each round, find the top corresponding object,
 - Either cuboids or Cylinder
- Results were improved
 - Minimum correspondence required;
 - Removal of fitted points



```

2014 for cand in None:
2015     if cand is None:
2016         if cand.find == "wall left" and best_wall_corr:
2017             if cand == correspond0 & cand_min < best_wall:
2018                 best_wall_corr = cand_corr
2019                 continue
2020             if cand == correspond1 and best_col_corr:
2021                 continue
2022             if cand == correspond0 & cand_min > best_column & best_col_corr:
2023                 continue
2024             chosen = cand
2025             break
2026         if chosen is None:
2027             turn_info["status"] = "quality collapse"
2028             turn_info["turn_info"] = turn_info
2029             break
2030         if chosen.find == "wall best":
2031             best_wall_corr = max(best_wall_corr, chosen == correspond0)
2032         else:
2033             best_col_corr = max(best_col_corr, chosen == correspond0)
2034     chosen_candidates = list(chosen_candidates) + [
2035         ("turn" + i + "active center is in before leftover points"
2036          + ("in_fit left" + len(hypotheses) + "hypotheses, confirmed 1",
2037           correspond0 & turn_info["turn_info"] > 10, "cm",
2038           ("in_fit left" + len(hypotheses) + "hypotheses, confirmed 1",
2039            correspond0 & turn_info["turn_info"] > 100, "cm",
2040            ("in_fit left" + len(hypotheses) + "hypotheses, confirmed 1",
2041             correspond0 & turn_info["turn_info"] > 1000, "cm",
2042             chosen),
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2281         (chosen),
2282         (chosen),
2283        
```

```
>Python ransac_primitive_fit.py
```



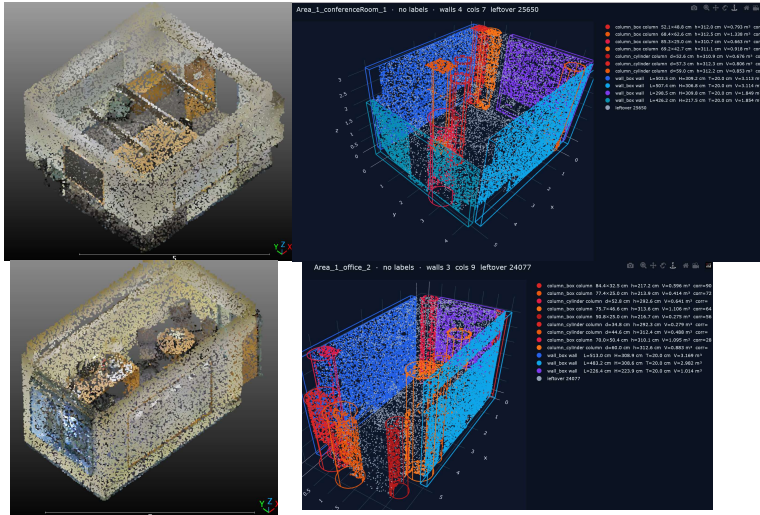

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III.2 Issues in

>Python s3dis_room_eval.py



- Columns from noises
- Columns fitting to doors, cabinets
- → semantic segmentation and mapping



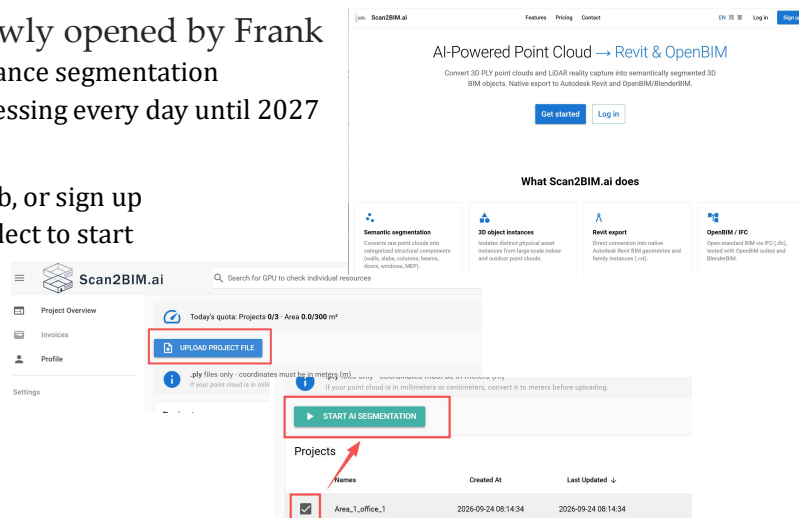
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III.3 Apply segmentation at <https://scan2bim.ai>

- Scan2bim.ai is newly opened by Frank
 - Semantic and instance segmentation
 - Free 300 m2 processing every day until 2027
- Log in
 - Use Google/GitHub, or sign up
 - Then upload → select to start





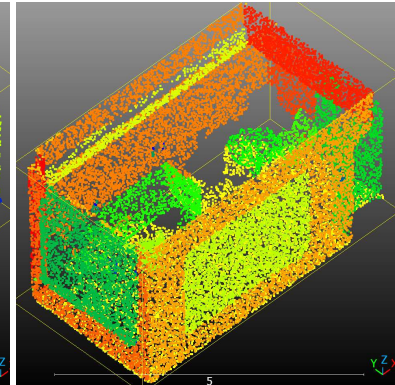
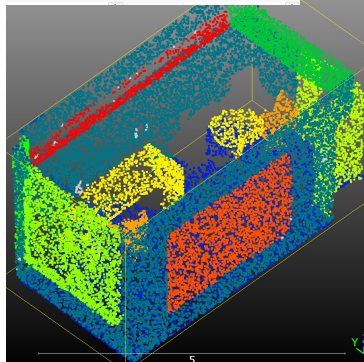
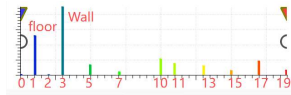
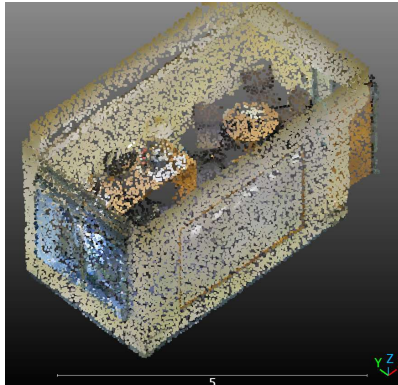
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III.3 Seg results

- Download segmented
 - Click PCD file name



Projects

Names	Created At	Last Updated	Status
☒ Area_1_office_1	2026-09-24 08:14:34	2026-09-24 08:16:08	Completed

Area_1_office_1 2026-09-24 08:14:34 2026-09-24 08:16:08 completed

Original files are deleted after processing completes. Segmentation results are retained for 7 days - please download them promptly

- ☒ Waiting in line for server to pick
- ☒ Processing
- ☒ Completed

DOWNLOAD

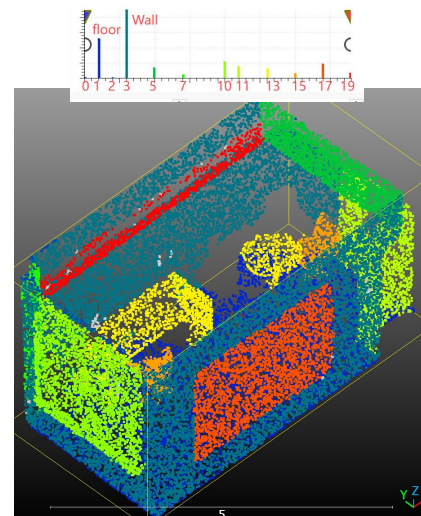


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III.3 Reference semantic label





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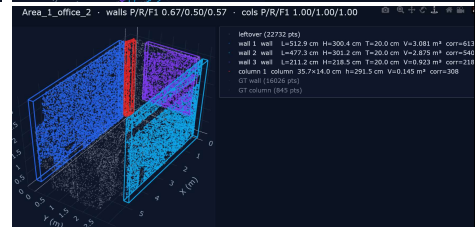
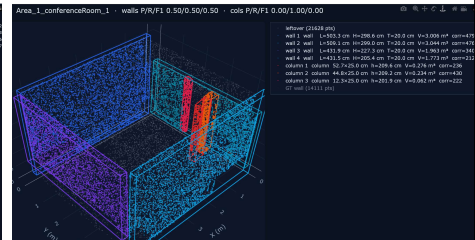
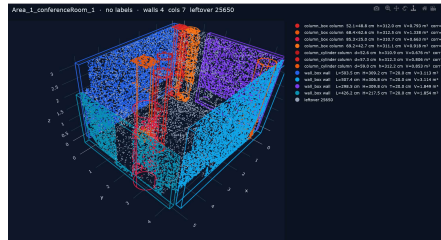
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III.3 Reference semantic label

- Noisy columns controlled. >Python s3dis_room_eval.py
– Score up
- Not enclosed...

	Precision	Recall	F1	3D IoU
Wall	0.74	0.56	0.63	0.48
Column (6 rooms with GT columns)	0.29	0.500	0.34	0.20
mIoU (wall, column)			0.48	0.34

	Precision	Recall	F1	3D IoU
Wall	0.767	0.533	0.623	0.497
Column (6 rooms with GT columns)	0.500	0.500	0.500	0.335
mIoU (wall, column)			0.561	0.438



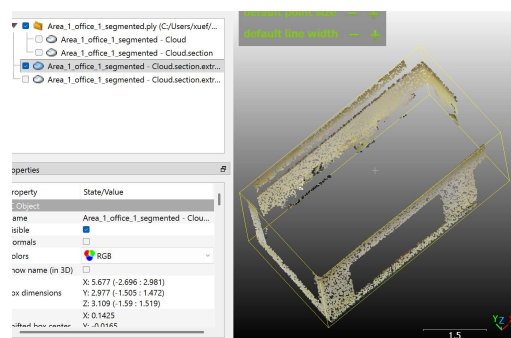
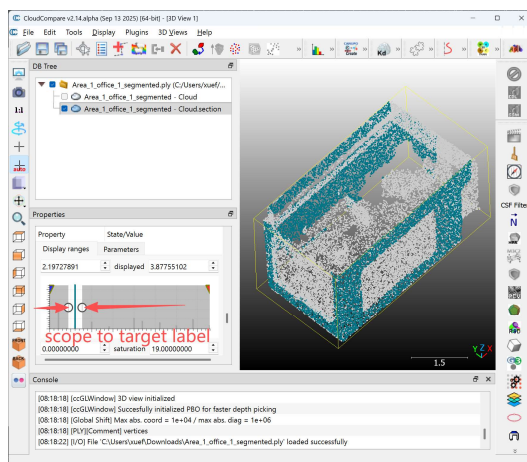
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III.4 How to manual proceed with semantic labels

- Then “Edit – Scalar – Filter by value”
– Only label = 3 is segmented.





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Part IV. Deep Learning training demo

(L2)



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IV.1 DL training for your own model

- How scan2bim.ai works
 - Semantic segmentation + instance segmentation
- Sem. Seg.
 - Assign each point a class.
 - Metrics: mIoU, accuracy (precision, recall).
 - $F1\ accuracy = 2 * precision * recall / (precision + recall)$
- Inst. Seg.
 - Also group each object (wall side, chair) into one.
 - Metrics: AP @ 0.25 / 0.50, mIoU_{3D}.
- Why training DL of semantic/instance segmentation
 - CUDA + pytorch + data
 - Installing files > 6 GB
 - Training > 30 mins
 - My laptop is weak (8GB)
 - And old structure, not supporting latest CUDA



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IV.1 Indoor semantic + instance open dataset

- S3DIS was released by Stanford in 2016
 - Source: Photogrammetry + manual labelling
 - Columns: Xyz *3, Color *3, semantic *1, instance *1

3D Semantic Parsing of Large-Scale Indoor Spaces

Iro Armeni¹ Ozan Sener^{1,2} Amir R. Zamir¹ Helen Jiang¹
Ioannis Brilakis³ Martin Fischer¹ Silvio Savarese¹

¹ Stanford University ² Cornell University ³ University of Cambridge

<http://buildingparser.stanford.edu/>

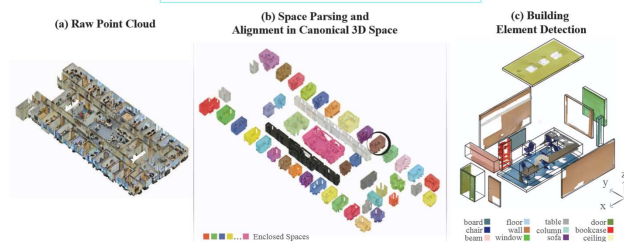


Figure 1: Semantic parsing of a large-scale point cloud. Left: the raw point cloud. Middle: the results of parsing the point cloud into disjoint spaces (i.e. the floor plan). Right: the results of parsing a detected room (marked with the black circle) into semantic elements.

https://pytorch-geometric.readthedocs.io/en/2.5.1/generated/torch_geometric_datasets.S3DIS.html



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IV.2 Well-known DL architecture/models

PointNet

2017

Per-point MLP + global max-pool. Permutation invariant. Misses local geometry. The only model we might fine-tune live on 8 GB.

KPConv

2019

Kernel points on the surface. Strong indoor baseline. Not a live-retrain in this room.

Point Transformer

2021–

Attention over points / neighbours. Heavy. Start from published weights.

Mask3D

2023

Instance head on a 3D backbone. Emits object masks — the IDs you can count. GitHub + authors' weights only today.

IV.2 What can a 8GB GPU run

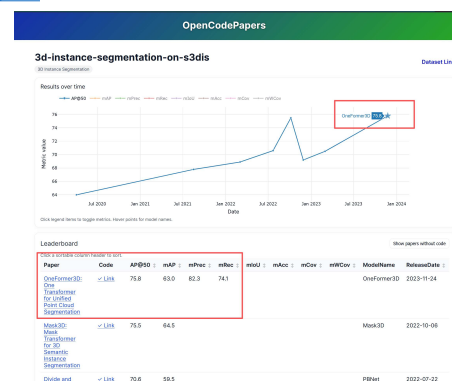
- Only the 2017 PointNet; pre-trained checkpoints

PointNet	charlesq34/pointnet · yanx27/Pointnet_Pointnet2_pytorch
KPConv	HuguesTHOMAS/KPConv · official PyTorch ports
Point Transformer	Authors' repos for the version you cite — check the year
Mask3D	JonasSchult/Mask3D — indoor instance, pretrained ScanNet weights



IV.3 How to start

- We have a 50 room small set at
 - <https://www.dropbox.com/scl/fo/n2hrhtpxyy8ovigch34wt/AKBM0425fH8SEf570s70oYk?rlkey=5h62f7ya118r3xjl0n2vf7afg&dl=0>
 - “>python pointnet_s3dis_train.py”
 - Then wait for some 40 mins.
- Full training data and benchmark at
 - <https://opencodepapers-b7572d.gitlab.io/benchmarks/3d-instance-segmentation-on-s3dis.html>





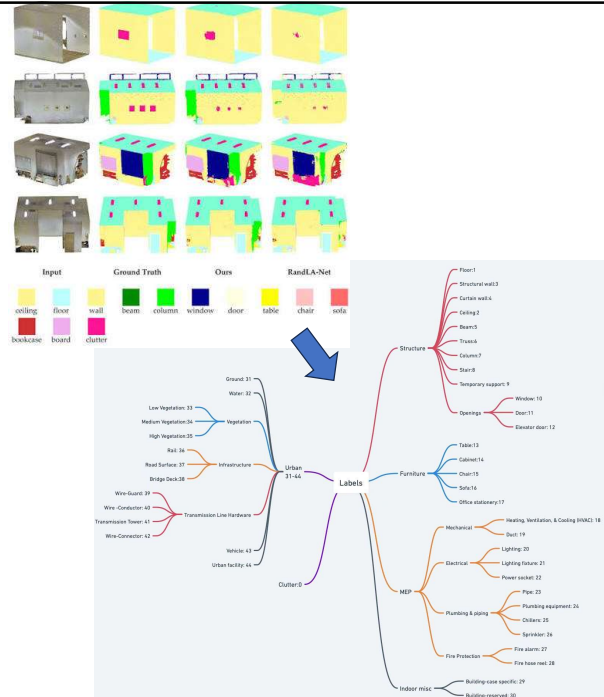
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IV.3 How to adapt to QS?

- No clue from Frank
 - perhaps add “Unit Rate” or “Relative weighting”?
- You can also extend the limited “labels”
 - We changed the 14 labels to 45 labels
 - To cover all building systems, and a few city objects
- How to do that
 - CloudCompare – Edit – Scaler – Update/Arithmetic



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IV.3 What yours may achieve, with extra cases

- Below were scan2bim.ai's performance

Pipeline scores (office, public housing)

F1 general AEs
90.48%

F1 BIM objects
72.45%

3D BIM object detection
61.21%

Worked example — chairs in office

Point-level:
mIoU 0.830
AP@0.25 0.984
AP@0.50 0.995

Instance level:
Inst. AP (all) 0.587
AP50 detect 0.731
AP25 detect 0.795

Library hook

CIC BIM object files
3,314

Family types
8,248

Indexed .rfa
3,129



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IV.4 Steps from dumb points to smart quantity

Direct outcome from: PCD – Semantic – instance pipeline

“An instance ID ____ of points grants us a countable object. A semantic label ____ only buys us a type. We cannot price a type.”

Final expected

“One payment item we could defend from an instance in this office/building is: _____, unit _____, exclusion _____.”



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Wrap-up

- Point cloud
 - Unstructured, surface data
 - Includes useful evidence to dig
- Point cloud editing
 - Basics, can segment levels and planes and objects
 - For instance and measurement
- Auto 3D clustering via RANSAC
 - Not very accurate; sensitive to indoor furniture and noise
 - Semantic/instance segmentation can improve it
- Training of DL models
 - Can extend for new QS target; But expensive, slow, requires data annotation





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Keep awesome !